```
#ifndef SEARCHINGALGO_H
#define SEARCHINGALGO_H

#include <iostream>

// PROBLEM 2
// Count the number of occurrences of a value
int countOccurrences(int arr[], int size, int target) {
    int count = 0;

    for (int i = 0; i < size; i++) {
        if (arr[i] == target) {
            count++;
        }
    }

    return count;
}


// PROBLEM 3
// Iterative Binary Search
void binarySearchDemo(int arr[], int size, int target) {
    int left = 0;
    int right = size - 1;

    while (left <= right) {
        int middle = left + (right - left) / 2;

        if (arr[middle] == target) {
            std::cout << "Searching is successful\n";
            return;
        }
        else if (target < arr[middle]) {
            right = middle - 1;
        }
        else {
            left = middle + 1;
        }
    }

    std::cout << "Searching is unsuccessful\n";
}


// PROBLEM 4
// Recursive Binary Search
bool binarySearchRecursive(int arr[], int left, int right, int target) {

    // Base case: value was not found
    if (left > right) {
        return false;
    }

    int middle = left + (right - left) / 2;

    // Value found
    if (arr[middle] == target) {
        return true;
```

```
    }

    // Search the left side
    if (target < arr[middle]) {
        return binarySearchRecursive(arr, left, middle - 1, target);
    }

    // Search the right side
    return binarySearchRecursive(arr, middle + 1, right, target);
}

#endif
```